Internship Defense

AI-Based Classification of HPV Status in Head & Neck Cancer Using CT/PET Radiomics and Clinical Data

Presented by:

Mohamad Ali El Anis El Dirany

Supervised by :

Dr. Mohamad hammoush

20/06/2026

Table of content

- Introduction
- Problem Statement & Objectives
- Internship Timeline
- State of the Art
- Methodology
- Results & Comparative Discussion
- Interpretability Insights
- Conclusion

Introduction

This internship was completed as part of a Bachelor's degree in Computer Networks & Telecommunications Engineering at the Lebanese University, Faculty of Technology, hosted by TOGETHERFORCHEIM in the field of Artificial Intelligence applied to medicine.

The project focused on classifying HPV status in Head & Neck Squamous Cell Carcinoma (HNSCC) patients using CT and PET radiomic features combined with clinical data. Because HPV status is a key prognostic biomarker but conventional testing is invasive and lab-dependent, the goal was to build a non-invasive, AI-based prediction tool to support clinical decision-making.



Together For Chehim & the CRPC

- Together For Chehim: non-profit association founded in 2015, based in Chehim, Lebanon
- CRPC (Collaborative Research and Projects Committee): established October 2023, a multidisciplinary academic & research platform
- Led by Dr. Nassib Abdallah (President) and Dr. Aya Hage Chehade (Vice-President)
- Four strategic pillars: academic training, international collaboration, access to education, continuous innovation
- 370+ hours of free, in-person training delivered locally
- Domains of expertise: AI & Data Science, Software Development, Electronics & Robotics, Industrial Systems, Languages & Soft Skills



Problem Statement & Motivation

Head & Neck Squamous Cell Carcinoma (HNSCC) is among the most common cancers worldwide, and HPV status strongly affects prognosis and treatment — HPV-positive tumors generally respond better to treatment and show improved survival outcomes. Yet conventional HPV testing relies on pathology and lab analysis, which is invasive and resource-intensive.

Artificial intelligence and radiomics can extract quantitative imaging biomarkers directly from CT and PET scans, and machine learning can detect complex patterns not easily visible to a human observer — motivating a non-invasive, AI-based HPV classification tool to support clinical decisions.



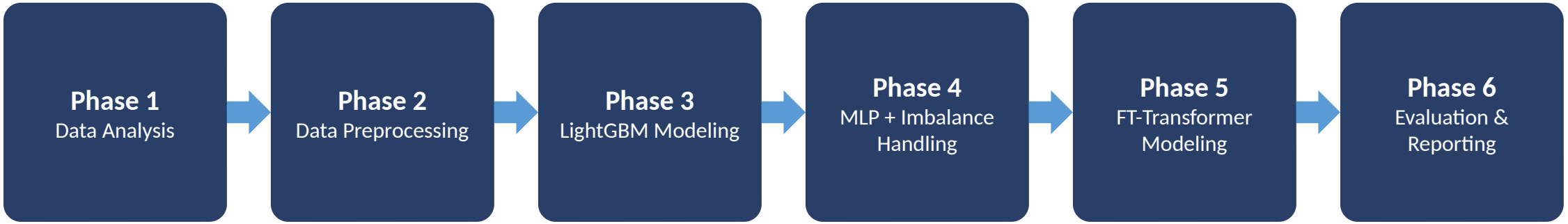
Objectives to Achieve

- Develop AI models to classify patients as HPV-positive or HPV-negative
- Compare three data sources: clinical-only, CT/PET radiomic-only, and combined clinical + radiomic
- Train and compare three model families: LightGBM, MLP, and FT-Transformer
- Evaluate models using Accuracy, ROC-AUC, Precision, Recall, F1, Balanced Accuracy, Specificity, PR-AUC, Brier Score and calibration
- Interpret model predictions through feature importance and SHAP analysis
- Identify the most reliable and clinically applicable approach



Internship Timeline

Six sequential phases, from dataset exploration to final model comparison and reporting



State of the Art — HPV, HNSCC & Imaging

High-risk HPV is a major cause of oropharyngeal head and neck cancer, and HPV status strongly influences prognosis and treatment. AI models commonly combine **PET/CT imaging** (metabolic and anatomical information) with **clinical data** using multimodal deep learning to predict HPV status non-invasively. The **HECKTOR 2025** challenge provides one of the largest multicenter FDG-PET/CT datasets (1,123 patients from 10 centers), enabling robust evaluation under real-world conditions.=

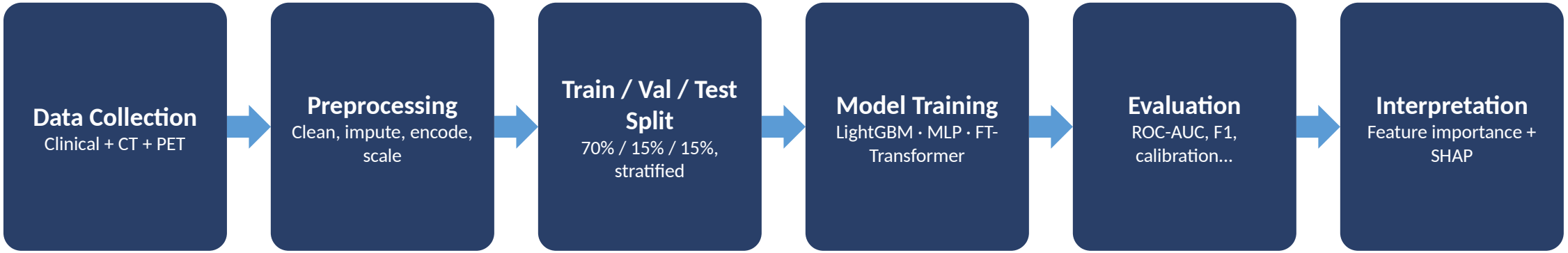
Key Takeaway

- PET/CT + clinical data is the current standard for AI-based HPV prediction.
- HECKTOR offers a realistic multicenter benchmark for evaluating robustness.
- Improving generalization and HPV-negative detection remains the main research challenge and motivation for this project.



Methodology

Consistent six-step pipeline applied identically to all three model architectures



Datasets Used

- **HECKTOR Clinical file**
  - Age, gender, tobacco/alcohol use, performance status, TNM staging, treatment, HPV status (target)
- **CT radiomics file**
  - First-order (intensity), texture (GLCM, GLRLM, GLSZM, GLDM, NGTDM), and shape descriptors (volume, surface area, sphericity)
- **PET radiomics file**
  - Equivalent radiomic feature families extracted from PET imagin

File	Number of Patients (Rows)	Number of Features (Columns, including `PatientID`)
HECKTOR_2025_Training_Task_3.csv	588	10
pet_features.csv	652	108
ct_features.csv	652	108



Data Preprocessing Pipeline

- Rule-based cleaning: invalid Performance Status values converted to missing (NaN)
- Missing-indicator columns flag originally-missing Tobacco / Alcohol / Performance Status values
- Most-frequent imputation for categorical clinical variables
- Constant imputation (“Unknown” category) for missing M-stage
- Min–Max scaling applied to Age
- One-hot encoding applied to the categorical M-stage variable
- Pipeline fitted on the training set only, then applied to validation/test — avoids data leakage



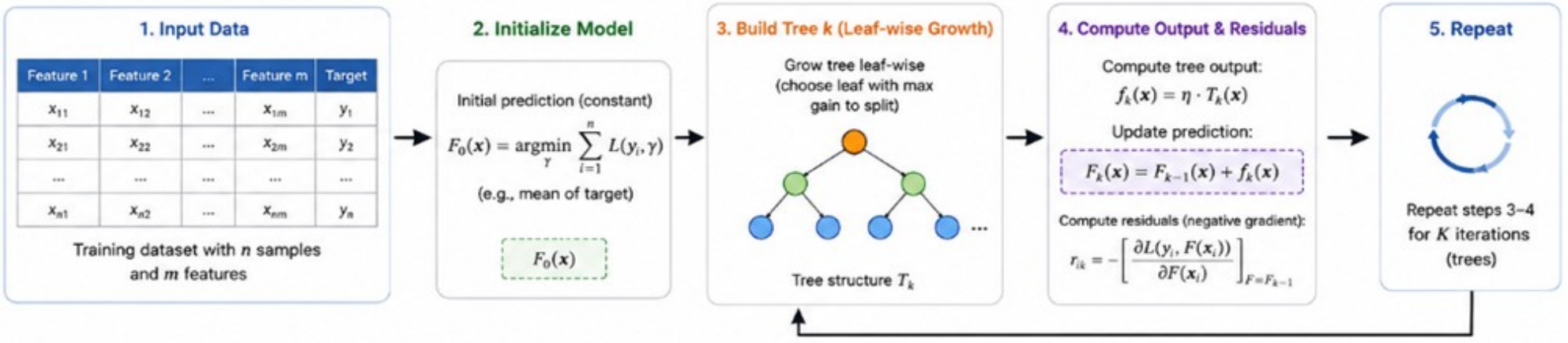
Data Split Strategy

- Clinical and PET data merged by Patient ID (inner join)
  - 74 clinical-only and 138 PET-only records dropped for lacking a match
- Split: 70% train / 15% validation / 15% test, stratified by HPV status
- Training set: 360 patients — 320 HPV-positive, 40 HPV-negative
- Validation set: 77 patients — 69 HPV-positive, 8 HPV-negative
- Test set: 77 patients — 68 HPV-positive, 9 HPV-negative



LightGBM Classification

- Gradient boosting framework: sequential decision trees, each correcting the errors of the previous ones
- Leaf-wise (rather than level-wise) tree growth — higher accuracy with fewer trees
- Uses GOSS and EFB for efficient training on high-dimensional radiomics data
- Well suited to medical classification: handles high-dimensional data, captures nonlinear relationships, provides feature importance
- Three experiments run: clinical-only, PET-only, and combined clinical + PET



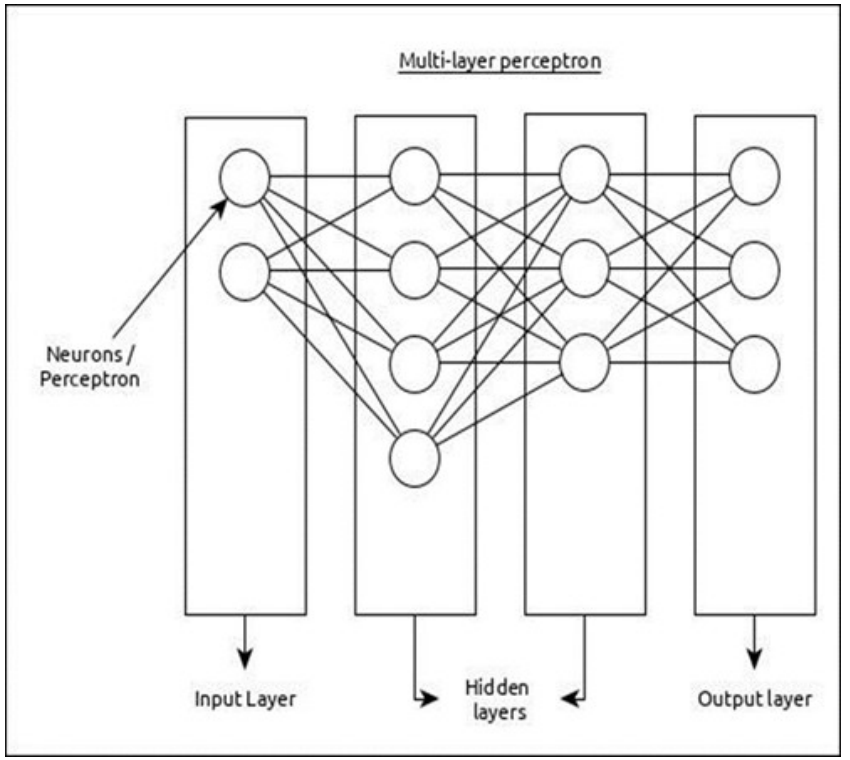
LightGBM Performance

Feature Set	Val AUC	Test AUC	Val Acc	Test Acc	Val Sens	Test Sens	Val Spec	Test Spec
Clinical-only	0.942	0.940	0.935	0.897	0.986	0.942	0.500	0.556
PET-only	0.900	0.549	0.896	0.872	0.986	0.971	0.125	0.111
Combined	0.975	0.857	0.922	0.936	0.986	0.986	0.375	0.556

The combined model gave the best ROC-AUC and accuracy; PET-only features failed to generalize to the test set.

MLP & Handling Class Imbalance

- Multi-Layer Perceptron: stacked dense layers with ReLU activations to learn nonlinear patterns
- Same protocol as LightGBM — identical splits, preprocessing, and metrics for a fair comparison
- Dataset is imbalanced (few HPV-negative patients), biasing the model toward the majority class
- Addressed with two oversampling techniques: SMOTE and SMOTENC (for mixed categorical/continuous data)
- PET-only features required an added StandardScaler step — without it, the model predicted every patient positive (0% specificity)
- Each configuration tested before and after balancing / scaling



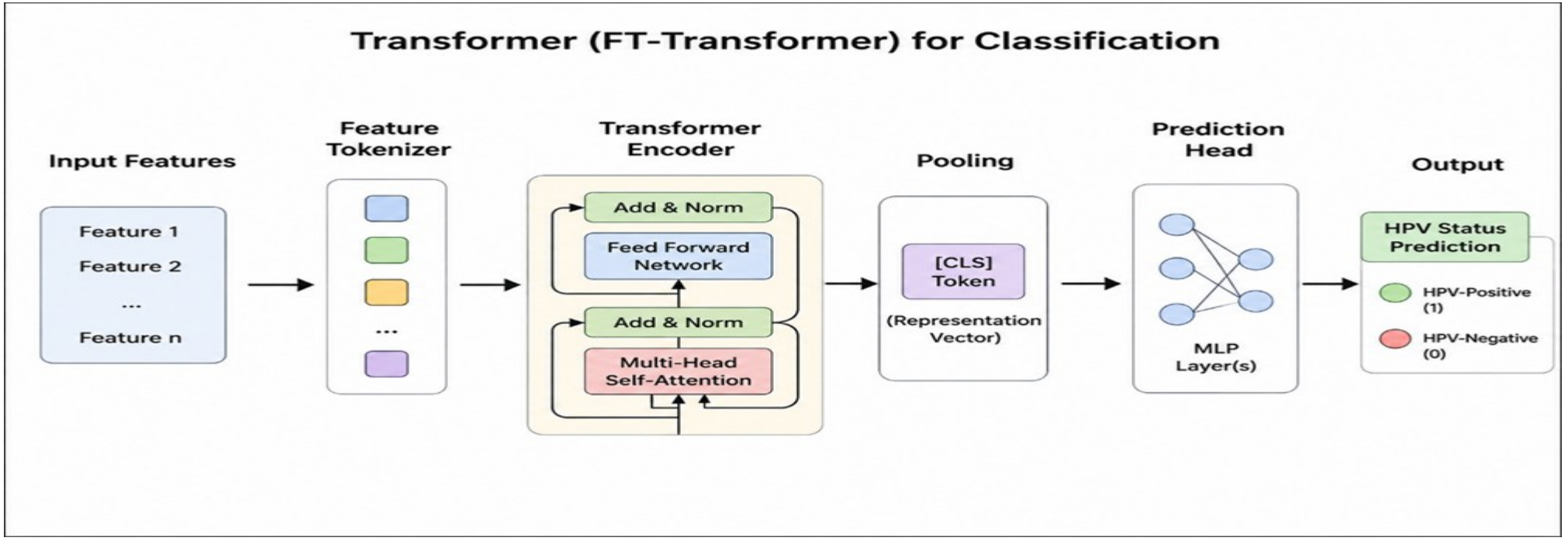
MLP Performance

Feature Set	Val AUC	Test AUC	Val Acc	Test Acc	Val Sens	Test Sens	Val Spec	Test Spec
Clinical-only	0.978	0.862	0.922	0.872	0.913	0.899	1.000	0.667
PET-only (scaled)	0.826	0.621	0.831	0.833	0.899	0.928	0.250	0.111
Combined (scaled)	0.976	0.813	0.935	0.910	0.986	0.957	0.500	0.556

SMOTE/SMOTENC and feature scaling were needed to make PET-only and combined MLP models usable; clinical-only remained the most stable.

FT-Transformer Classification

- Transformer-based deep learning architecture adapted for tabular data
- Learns feature interactions directly from clinical and radiomic tabular inputs
- Same training / validation / test protocol, with no extra tuning — for a fair three-way comparison
- Three experiments run: clinical-only, PET-only, and combined clinical + PET



FT-Transformer Performance

Feature Set	Test ROC-AUC	Test Accuracy	Test Sensitivity	Test Specificity
Clinical-only	0.820	0.936	0.986	0.500
PET-only	0.638	— (predicted all positive)	1.000	0.000
Combined	0.916	0.936	0.986	0.556

PET-only collapsed to predicting every patient positive (0% specificity); combining PET with clinical data restored balance and the best ROC-AUC.

Comparative Discussion

Model (Clinical + PET)	Test ROC-AUC	Test Accuracy	Test Sensitivity	Test Specificity
LightGBM	0.857	0.936	0.986	0.556
MLP	0.813	0.910	0.957	0.556
FT-Transformer	0.916	0.936	0.986	0.556

Across all three architectures, combining clinical and PET features produced the most balanced result; FT-Transformer gave the strongest overall test performance.

What Drove the Predictions

Across every model, M-stage was consistently the single most important feature, with Tobacco and Alcohol Consumption (and their missingness indicators) repeatedly ranking among the top predictors. PET contributed mainly shape and texture descriptors — Sphericity, Elongation, Maximum2DDiameterSlice, and gray-level features — but played a complementary rather than dominant role. SHAP analysis confirmed these permutation-importance rankings across LightGBM, MLP, and FT-Transformer.

Consistent Pattern Across Models

- Clinical features alone: strong, stable signal in every model
- PET features alone: high sensitivity but poor specificity, weak generalization to test data
- Clinical + PET combined: most balanced, best-calibrated, best-generalizing models overall



Conclusion

This internship developed and compared three model families across three feature sets for non-invasive HPV status classification in HNSCC. Clinical data was consistently the strongest and most reliable predictor, while PET radiomics alone were insufficient but added complementary value once combined with clinical data. The best overall result came from the FT-Transformer on combined clinical+PET features — test ROC-AUC 0.916, PR-AUC 0.988, accuracy 0.936, and sensitivity 0.986



Lebanese University
Faculty of Technology

Comité de
Recherche et
Projets
Collaboratifs

CHEHIM
TOGETHER FOR CHEHIM
Founded in 2015
A.D No. 1745

Internship Defense

Thank You

Questions & Discussion